Intersections in simultaneous group actions

Human Compression project

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Abstract

When a connected component of a product of two permutation actions is projected onto its first coordinate, an intersection can be repaired there before its remaining defect is tested in one fiber. This observation gives the path criterion for mixes of polytopes without a symmetry assumption. A normalizer hypothesis then identifies the fibers with actual groups of automorphisms. We separate this algebraic identification from the further transitivity needed to reduce the number of flags tested. The resulting applications are the mix criteria of Cunningham and Hubard; the elementary image-and-kernel argument, rather than a claim of historical novelty, is the common method.

1 Repairing the first coordinate

Let a group W act on the left on sets X and Y. It acts diagonally on $X \times Y$. For $S \leq W$ and $q = (x, y)$ write

$$Sq = \{(sx, sy) : s \in S\}, \qquad D_S(q) = \{y' : (x, y') \in Sq\} = (S \cap \mathrm{Stab}_W(x))y.$$

The last equality is literal: a word reaches a point with first coordinate x precisely when that word fixes x. No regularity, finiteness or normality is assumed.

Theorem 1. *Let $A, B, C \leq W$, with $C \leq A \cap B$, and fix $q = (x, y)$. Suppose that $Ax \cap Bx = Cx$. Then*

$$Aq \cap Bq = Cq \quad \Longleftrightarrow \quad D_A(q) \cap D_B(q) = D_C(q). \tag{1}$$

Proof. The forward implication follows by taking the points with first coordinate x in the asserted equality. Conversely let $q' = (x', y') \in Aq \cap Bq$. Its first coordinate belongs to $Ax \cap Bx$, so choose $c \in C$ with $x' = cx$. Then $c^{-1}q'$ belongs to both Aq and Bq, since c lies in both subgroups. Its first coordinate is x, so the equality of the three fibers gives $c^{-1}q' \in Cq$. Multiplying by c gives $q' \in Cq$. The opposite inclusion follows from $C \leq A \cap B$. □

This proof is the action version of a familiar subgroup argument. If two subgroups have the expected common image under a homomorphism, subtract a lift of that image and test only the remaining kernel. In (1), the first action replaces the quotient homomorphism. Stabilizers need not be normal, so the first formulation is in terms of sets rather than quotient groups.

2 Recursive path intersections

Put $I = \{0, \dots, n-1\}$ and let

$$W = W_n = \langle r_i \ (i \in I) \mid r_i^2 = 1, \ [r_i, r_j] = 1 \ (|i - j| > 1) \rangle.$$

For $J \subseteq I$, put $W_J = \langle r_j : j \in J \rangle$. A connected premaniplex is a transitive W-set. Its colored adjacencies are the actions of the r_i; loops and coincident differently colored edges are allowed. A maniplex additionally has no fixed point of any r_i or $r_i r_j$ with $i \neq j$. We retain the standard theorem

of Garza-Vargas and Hubard: a maniplex is the flag graph of a polytope exactly when its color paths have the intersection property. The interval form used below is

$$W_{[0,j]}x \cap W_{[i,n-1]}x = W_{[i,j]}x. \tag{2}$$

The corresponding cases with a single common color are included. In particular rank two maniplexes are polygons, finite or infinite, and satisfy the condition. This geometric characterization is an attributed input; it is not part of the algebra in Theorem 1.

Lemma 2. *For an n-maniplex Q, $n \geq 3$, the following are equivalent:*

1. *Q is polytopal;*
2. *its facets are polytopal and, at every flag q and every $0 \leq i \leq n-2$,*

$$W_{[0,n-2]}q \cap W_{[i,n-1]}q = W_{[i,n-2]}q;$$

3. *its facets and vertex-figures are polytopal and, at every flag q,*

$$W_{[0,n-2]}q \cap W_{[1,n-1]}q = W_{[1,n-2]}q. \tag{3}$$

Proof. Polytopality implies the other conditions by (2), including its application inside faces. To prove (2) from the second condition, take a flag q' in its left-hand side for arbitrary i, j. If $j = n-1$ there is nothing to prove. Otherwise the second condition puts q' in $W_{[i,n-2]}q$. Inside the common facet, its intersection with $W_{[0,j]}q$ is exactly $W_{[i,j]}q$, because that facet is polytopal.

The third condition implies the second. The case $i = 0$ is immediate. For $i \geq 1$, a flag in $W_{[0,n-2]}q \cap W_{[i,n-1]}q$ lies, by (3), in $W_{[1,n-2]}q$. Inside the vertex-figure, intersecting that orbit with $W_{[i,n-1]}q$ gives $W_{[i,n-2]}q$. This proves all assertions, using the retained path characterization only at the beginning and end. □

Let X be a polytope, Y a connected premaniplex, and let Q be the orbit of a chosen pair in $X \times Y$. This is their rooted mix. It is a maniplex: a fixed point of a generator or a product of two distinct generators would project to such a fixed point in X. Every restricted-color component of Q is the mix of the corresponding rooted components of X and Y.

Corollary 3. *Write $A = W_{[0,n-2]}$, $B = W_{[1,n-1]}$, $C = W_{[1,n-2]}$. Then Q is polytopal if and only if its facets and vertex-figures are polytopal and*

$$D_A(q) \cap D_B(q) = D_C(q) \quad \textit{for every flag } q \in Q. \tag{4}$$

Proof. The first factor is polytopal, so $Ax \cap Bx = Cx$ at every flag x. Apply Theorem 1 to (3) at every flag of the mix, then apply Lemma 2. □

There is no admissibility assumption in this criterion. It keeps the fiber as a set when there is no appropriate group of automorphisms.

3 When the fibers are groups

Fix base flags x, y and put $N_1 = \mathrm{Stab}_W(x)$, $N_2 = \mathrm{Stab}_W(y)$. The action on Y is the action on the left cosets W/N_2. If $u \in \mathrm{N}_W(N_2)$, the rule

$$(gN_2)[u] = guN_2$$

defines an automorphism commuting with the left W-action. Conversely, an automorphism commuting with W is determined by the image of N_2; that image is uN_2 for some u normalizing N_2. Thus the group

of these right automorphisms is $\mathrm{N}_W(N_2)/N_2$. Its action is free on flags: one fixed flag determines the identity by transitivity.

Suppose only that $N_1 \leq \mathrm{N}_W(N_2)$. For $S \leq W$ define the actual subgroup

$$V_S(q) = \{[u] \in \mathrm{N}_W(N_2)/N_2 \,:\, u \in N_1 \cap S\}. \tag{5}$$

The evaluation $[u] \mapsto y[u]$ identifies $V_S(q)$ bijectively with $D_S(q)$. This follows directly from the definitions; its injectivity uses the free action just proved. Consequently

$$D_A(q) \cap D_B(q) = D_C(q) \quad \Longleftrightarrow \quad V_A(q) \cap V_B(q) = V_C(q). \tag{6}$$

Here $V_C \leq V_A \cap V_B$, so inclusion in V_C is an equivalent test. This also repairs the proof of [CH, Proposition 5.2]. Normality of the denominator in the product is sufficient: if $u \in N_2$ and $v \in N_1$, then $uvN_2 = vN_2$ since v normalizes N_2. No reverse normalization is inferred from this quotient. At another flag gq, both stabilizers are conjugated by g, and the normalizing hypothesis persists. Formulae (5)–(6) therefore apply at every flag; they must be formed at that flag, not just at the original base.

Lemma 4. *Under the normalizing hypothesis, the lower variance group of the S-components in Y relative to those in X is well-defined and embeds in* $\mathrm{Aut}(Y)$ *with image* $V_S(q)$.

Proof. The two component stabilizers in S are $N_i \cap S$. Since N_1 normalizes N_2, $N_1 \cap S$ normalizes $N_2 \cap S$. The lower variance group is

$$(N_2 \cap S)(N_1 \cap S)/(N_2 \cap S).$$

Sending the class of $u \in N_1 \cap S$ to its class modulo N_2 is well-defined and injective: its kernel is exactly $N_1 \cap N_2 \cap S$. Its image is (5). This proves the actual embedding rather than merely an abstract isomorphism of the variance groups. In particular the intersection in (6) takes place inside one specified automorphism group. □

The same quotient calculation supplies the upper variance group. Under $N_1 \leq \mathrm{N}_W(N_2)$, the map

$$N_1/(N_1 \cap N_2) \longrightarrow N_2N_1/N_2, \qquad u(N_1 \cap N_2) \longmapsto uN_2$$

is a well-defined surjective homomorphism with trivial kernel. It is therefore an isomorphism, with no cardinality argument. The group is trivial exactly when $N_1 \leq N_2$, which is exactly a rooted covering $X \to Y$. If instead $N_2 \leq N_1$ and the quotient is defined, both variance groups are N_1/N_2. These statements include the source's covering examples and Proposition 5.4. For regular inputs the upper group is the kernel of the projection of their mixed marked groups, and the lower group is the kernel of the projection to their largest common quotient $W/(N_1N_2)$.

For finite polygons with p and q sides, the two stabilizers in the infinite dihedral group are the cyclic groups generated by the pth and qth powers of its rotation. Their intersection is generated by the least-common-multiple power, and their product by the greatest-common- divisor power. This proves the source's Example 5.6, including lower variance order $q/\gcd(p,q)$ for $X(M_2 \mid M_1)$ when M_1 is the p-gon and M_2 the q-gon. Reversing the arguments gives $p/\gcd(p,q)$, exactly as printed there. For an infinite polygon its stabilizer is trivial, so the same assertions are read directly as subgroup statements rather than as arithmetic with an undefined infinite greatest common divisor.

If N_1, N_2 are both normal in a common subgroup $V \leq W$, then each normalizes the other. The premaniplex W/V is a common symmetry quotient, and the terminology of Cunningham–Hubard is that both inputs are T-admissible for $T = W/V$. The stabilizer of the mix is $N = N_1 \cap N_2$, and V/N acts by right automorphisms on W/N. The diagonal map $V/N \to V/N_1 \times V/N_2$ is injective, since its kernel is precisely N. Each restricted component has the same property with $V \cap S$ in place of V. These facts recover the admissibility and subgroup identifications used in the source's main criterion.

The restriction statements of [CH, Proposition 2.6] follow as well. For orientability, a coloring is a character ϵ_I vanishing on the flag stabilizer. Restrict that character and stabilizer to W_J; if the character becomes zero, use the one-color coloring. This is an explicit boundary convention for Proposition 2.6: the restricted quotient is the connected one-flag component. We do not assert a surjective covering of two disconnected flags with only semi-edges when the restricted index set contains every color. For the rotary assertion, a regular component remains regular. Otherwise the input has two flag orbits of class 2_I. Its flag stabilizer is normal in $V = \ker \epsilon_I$, so $N \cap W_J$ is normal in $V \cap W_J$. The latter subgroup has index at most two in W_J. Consequently the component is regular or has two flag orbits with precisely the restricted color character. If a larger normalizer merges those two orbits, it is regular, which is still rotary. Combining these two arguments gives the admissibility assertion, for facets and vertex-figures with the usual renumbering of their colors.

4 How many flags must be tested?

At this point there are two different reductions: (6) replaces fibers by groups; a further orbit argument may reduce the number of flags at which (6) is tested. They have different hypotheses.

Proposition 5. *Let V contain and normalize N_1, N_2. One representative of each double coset in $C\backslash W/V$ suffices for the test (4). In particular one base flag suffices if $W = CV$.*

Proof. For $c \in C$, multiplication by c fixes each of the three orbit sets Aq, Bq, Cq as sets, because $C \leq A \cap B$. Thus equality (3) at q is equivalent to equality at cq. Right automorphisms induced by V/N preserve all three colored path relations. If $g' = cgv$, then $g'N = cgvN$ is obtained from gN by such a right automorphism and a left multiplication by c. The condition is therefore constant on each double coset. Corollary 3 proves the assertion. □

The underlying component argument also holds for any set J of colors: if $W = W_J V$, the right action of V/N is transitive on the W_J-components of the mix. Indeed $g = jv$ places gN in the component obtained from the base component by v. With J the complement of a chain type, these are the chain components used in [CH, Proposition 4.6]. This proves its arbitrary-color assertion, not only the middle-color case needed below.

The condition $W = CV$ says exactly that the common quotient $T = W/V$ is connected on the middle colors. More generally, its C-components index the displayed sufficient list of tests. This assertion does not infer transitivity of a mix from transitivity of its two factors.

There is a further useful reduction when the mix has automorphisms which do not project to the two factors. Condition (3) itself is invariant under every automorphism of the mix and under middle-color paths. Thus one base flag suffices whenever the mix is medial-section- transitive. In particular it suffices when the mix is regular, regardless of whether its full automorphism group acts coordinatewise. This uses only the path equality, followed by its already proved equivalence with the fiber or variance test at that base.

For regular inputs take $V = W$, so there is one test. For two-orbit inputs in a fixed class, take V to be the kernel of the character $\epsilon_I : W \to C_2$ with $\epsilon_I(r_i) = 0$ for $i \in I$ and 1 otherwise. If some middle color is outside I, then $CV = W$. If all middle colors lie in I, there are two tests, represented by 1 and any r_j with $j \notin I$. The possible complements of I are then $\{0\}, \{n-1\}, \{0, n-1\}$. Applying the same argument with A and B describes whether there is one orbit or two of facets and vertex-figures. This gives the three exceptional tables in the application without silently identifying different rooted sections.

5 A useful quotient consequence

The variance group in (5) is trivial exactly when every word of S fixing the first base flag also fixes the second. Equivalently the first rooted S-component covers the second. Thus a covering relation

between facets makes the facet variance trivial.

Proposition 6. *Suppose the two inputs are admissible over a common T, the first is polytopal, and at every paired flag its first facet covers its second facet. Then their mix is polytopal.*

Proof. Proceed by induction on the rank, starting with polygons. The mix of a polygon and a rank-two premaniplex is a rank-two maniplex, hence a polygon. In higher rank the mixed facets are the first facets and are polytopal. The facets of each mixed vertex-figure are mixes of paired medial sections. The assumed facet coverings restrict to coverings between these sections. Admissibility restricts to the vertex-figures, so induction shows that all mixed vertex-figures are polytopal. Finally $V_A(q) = 1$ at every flag, so (6) and Corollary 3 give polytopality of the whole mix. Interchanging the two ends proves the dual result. □

6 Attribution and remaining application checks

The path characterization is retained from Garza-Vargas–Hubard. In the source's symmetry-type discussion on physical page 4, adjacency between two flag orbits means that each flag has its unique neighbor in the other orbit, not that it is adjacent to every flag there. This is immediate because automorphisms commute with the color involutions. The mix, admissibility and variance terminology, the recursive criteria and the principal polytopality theorem are due to Cunningham–Hubard. The first-coordinate proof above is their correction-of-a-path argument expressed for arbitrary group actions; it is also the same elementary image-and-kernel mechanism used in our supported permutation proofs. The one-sided normalizer formulation states exactly the algebra needed. We make no historical novelty claim for it.

The companion application binds the printed single-base corollaries to a justified orbit hypothesis and exhibits their failing literal special cases. Its source table separately identifies the criteria in which actual roots replace suppressed identifications. Nothing here proves any of the source's three concluding open problems. This working edition carries no source count, conjecture count or numerical comparison.

References

[1] G. Cunningham and I. Hubard, *Polytopality criteria for the mix of polytopes and maniplexes*, arXiv:2506.19334v1, 24 June 2025.

[2] J. Garza-Vargas and I. Hubard, *Polytopality of maniplexes*, Discrete Math. 341 (2018), 2068–2079, Theorem 5.3; retained through [CH, Theorem 2.1].